Rank Fusion and Learning to Rank - Report

COL 764/7364 Assignment 3 · Maaz Akhtar · 2023BB10156 · Submission #597 (V12)

1. Final system, models and data use

This report describes **submission #597**, the evaluated V12 archive. It implements handwritten rank fusion, pointwise and pairwise neural training, a LightGBM LambdaMART-family scorer, and a portable training recipe. Later experimental scorers are excluded.

Fusion. The submitted method is per-query min-max **CombSUM**, with weights 0.3 for BM25 whole-document, 0.3 for BM25 title, 0.4 for LMIR.DIR and 0 for LMIR.JM. RRF and CombMNZ are also implemented. A fixed 1,146-configuration TRAIN-only comparison selected these weights; no DEV labels enter fusion tuning.

Neural models. Pointwise training minimizes mean squared error. RankNet uses the stable loss **mean[softplus(other score - preferred score)]**; sampled pairs stay within a query, exclude equal grades and put the preferred document first. The competitive pairwise trainer additionally weights pairs by their nDCG swap effect. Each member has 136 → 128 → 64 → 1 ReLU layers, 30 MSE warm-up epochs at learning rate 0.001 and two RankNet epochs at 0.0001, with fresh Adam, batch 1,024 and at most 300 pairs/query/epoch. Seeds 0 and 41 are equally averaged and compiled into a staff-loadable 136 → 256 → 128 → 1 MLP. Pointwise uses six MSE epochs.

Tree features and training. Six weighted LambdaRank members form one native 3,800-tree model. The 136 provided features are supplemented by query-local midrank percentiles and min-max values, giving 408 internal features. Transforms use only current candidates, preserve missing values, and map constants to 0.5. Training-only copies of queries with affine changes, feature dropout and noise encourage robustness; they do not recreate the hidden C2 versions. See Section 4 for exact member settings [1,2].

Split declaration. The tree was fitted on released Fold 1 **TRAIN+DEV**: 723,412 + 235,259 rows, 6,000 + 2,000 queries, with split-namespaced query IDs. Its DEV score is **in-sample**. Tree hyperparameters/weights used grouped TRAIN-only CV; final counts were fixed with no early stopping. Both neural artifacts were fitted on TRAIN only, with standardization learned from training rows. Pointwise epoch selection used internal TRAIN fold 0 with patience 5, then a six-epoch full-TRAIN refit; pairwise used fixed-epoch five-fold selection. No neural fitting or stopping used released DEV.

2. Recorded results for the submitted system

Model / method	DEV nDCG@10	Interpretation
Single BM25 feature	0.288487	Untrained reference
Pointwise MLP	0.431708	TRAIN-only fit
Pairwise MLP	0.452027	TRAIN-only fit
LambdaMART ensemble	**0.682399**	**TRAIN+DEV fit; in-sample**

Fusion achieved **0.3090 nDCG@10 / 0.1297 MAP@10**. The supplied #597 grader snapshot reports **P = 0.4419** on held-out queries, **C2 mean = 0.3795**, 7/7 conformance and 10/10 correctness. Its checkpoint position was 10/35, live score 0.4722/0.65. These class-relative values are a recorded checkpoint, not a final rank; no C1 score was supplied.

3. Correctness, runtime and transfer recipe

The V12 release passed **176 tests** locally, on the college server and in CI, including loss/SGD examples, pair sampling, determinism, staff loading and interface checks. Its two fresh pairwise fits were byte-identical; the tree was the retained, previously verified V10 artifact, not a new V12 fit. Recorded preparation was **5.680 s**, worst fusion **0.0011 s** and reranking **0.4885 s**, inside the 60 s / 2 s limits. Local staff loading used Python 3.12.14; HPC used Python 3.9.21.

The unchanged C1 recipe accepts arbitrary feature widths, uses raw values plus query percentiles, and chooses between two leaf capacities from query count (15/31, 31/63 or 63/127). Its supplied validation file selects capacity/iteration; it uses learning rate 0.05, L2 = 10, patience 40 and a deterministic 200-500-tree budget. No hidden benchmark was identified or accessed.

4. Tree-count sweep, generalization and final configuration

Controlled tree-count comparison. The archived TRAIN-only sweep varied 200/400/800 trees in an unaugmented query-rank member, holding the other settings fixed (127 leaves, minimum leaf 100, L2 = 10, learning rate 0.05). The corresponding DEV prefix scores below were computed **after submission for this report, on 24 October 2026**, from that member inside the exact #597 artifact. They are retrospective **in-sample** measurements, not new tuning or replacements for the submitted model.

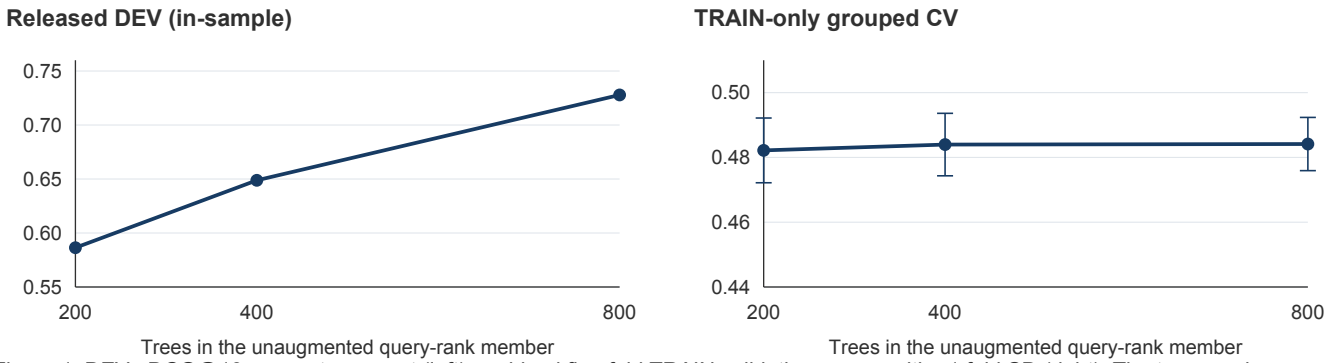


Figure 1. DEV nDCG@10 versus tree count (left); archived five-fold TRAIN validation means with ±1 fold SD (right). The two panels use different vertical scales. The member's positive export weight changes scores but not its ordering; these are component scores, not the full six-member ensemble.

Trees	DEV (in-sample)	CV train	CV validation mean ± SD	Public diagnostic mean
200	0.586381	0.618379	0.482170 ± 0.009994	0.460040
400	0.648799	0.694359	0.483968 ± 0.009643	0.459174
800	0.727855	0.782033	0.484133 ± 0.008215	0.454990

Observed gap. From 200 to 800 trees, the component's in-sample DEV rises from 0.586381 to 0.727855, while grouped-CV validation rises only from 0.482170 to 0.484133; the public diagnostic mean falls from 0.460040 to 0.454990. Thus extra fitting capacity mostly improves seen-query performance. For the final ensemble's TRAIN-only CV counterpart, mean training nDCG is **0.732114**, unseen-fold mean **0.487560 ± 0.009312**, and validation/training ratio **0.665962** (absolute gap 0.244554). This is a method-level estimate on TRAIN, not CV of the later TRAIN+DEV refit.

All five folds are grouped by query (seed 0); validation queries never contribute fitting rows or standardizer statistics. Counts are fixed rather than stopped on the reported fold. Regularization combines minimum leaf size 100, L2 = 10, row/feature subsampling, query-relative features, augmentation and averaging. The final TRAIN-only diagnostic means are affine 0.479000, dropout 0.477154 and noise 0.474921 (mean 0.477025). These public perturbations are diagnostics, not estimates of staff C2.

Final configuration and tradeoff. I retained the six-member tree below. All use LambdaRank, 127 leaves (no separate depth cap), learning rate 0.05, minimum leaf 100, L2 = 10, feature and row fractions 0.8, bagging frequency 1, seed 0 and four threads. The fifth and sixth members differ through extra-tree randomization; the sixth enables extra trees. Truncation is 30 except member 3 uses 13. The native export preserves the weighted sum; the full model's 0.682399 DEV score is not the 800-tree component's score.

Member features	Trees	Weight	Training augmentation
1. Raw + percentile	800	0.0703125	None
2. Min-max + percentile	200	0.0703125	None
3. Raw + percentile	800	0.1406250	Half-query dropout / noise
4. Raw + percentile + min-max	800	0.2812500	Half-query mixed perturbations
5. Raw + percentile + min-max	800	0.1875000	Full-query mixed perturbations
6. Same features; extra trees	400	0.2500000	Full-query mixed perturbations

V12's four new tree candidates / 33 settings did not pass the frozen robustness gates (mixture gain 0.000311 < 0.0005; diagnostic gain 0.000551 < 0.001), so the evaluated tree bytes were retained. For pairwise training, 50 fresh CV member fits favored 30-epoch warm-up over the 20-epoch control: mean 0.441474 versus 0.440371, gain 0.001103 with four positive folds. The fresh control reproduced archived evidence before promotion. Choices were committed before final fitting/DEV evaluation. I favored supported CV/robustness gains over accepting every marginal increase in seen-query accuracy.

What the grader adds. The pairwise DEV-to-held-out difference is 0.4520 - 0.4419 = 0.0101. Tree DEV 0.6824 versus C2 0.3795 shows why the large DEV score cannot establish generalization: DEV was in fitting and C2 averages different hidden conditions. Their difference is not a matched clean-test overfitting estimate. No per-version C2 results or C1 outcome were available.

5. AI-use disclosure and code provenance

AI use. I directed the development, set experimental priorities and chose submission iterations. I used **OpenAI Codex** as a programming assistant to help read the specification, propose and implement methods, write and revise code, test, evaluate, debug, execute college-HPC experiments, interpret results, package submissions and prepare this report. I uploaded the archives and checked the grader outcomes. The assistance also covered repetitive coding tasks and documentation.

File provenance. The table groups principal files/families in the V12 archive and the later report files. *Assignment-specific* means created with the AI assistance above. The course harness, network/standardizer, metrics, grouped-fold helper and interfaces supplied the scaffolding. No external fusion library, ready-made neural ranking loss/trainer, another student's solution or extra relevance-label dataset was used. The tree learner is the permitted LightGBM implementation [1,2].

Edited / added file(s) or family	What it does	Source / provenance
submission/ranking.py	Preparation, candidate-only fusion/reranking; stable tie handling.	Course interface; assignment-specific internals.
submission/fusion.py submission/fusion_config.json	Handwritten RRF, CombSUM/MNZ and weighted variants; selected weights.	Assignment formulas / primer; TRAIN-selected settings.
submission/pointwise_pairwise.py	MSE, pairs, stable RankNet, SGD steps, weighted training and seeded initialization.	Course skeleton extended; handwritten loss/training logic [1].
submission/data.py submission/features.py	Bounded SVMLight loading; raw, percentile and min-max representations.	Assignment-specific; course data/interface conventions.
submission/tree_ensemble.py scripts/pairwise_ensemble.py	Compile weighted trees and two neural members into standard loadable artifacts.	Assignment-specific numeric export; LightGBM/PyTorch formats.
submission/recipe.py	Adaptive, budgeted training for staff-supplied benchmark files.	Assignment-specific policy; LightGBM learner [2].
submission/model.txt	Retained V10 six-member, 3,800-tree scorer.	LightGBM 4.6.0; released TRAIN+DEV only.
submission/pointwise_model.pt submission/pairwise_model.pt	Six-epoch pointwise fit; V12 two-seed pairwise fit.	PyTorch; TRAIN only. Pairwise independently reproduced.
scripts/train_model.py scripts/cv_lambdamart*.py scripts/lambdamart_utils.py	Tree fitting, fixed-count grouped CV and native export checks.	Assignment-specific / adapted entry points; course CV/metrics.
scripts/cv_pairwise*.py scripts/augmentation.py scripts/robustness.py	Neural comparisons, query augmentation and label-preserving stress diagnostics.	Assignment-specific; public diagnostic assumptions.
scripts/evaluate_models.py scripts/record_artifacts.py scripts/hpc_*.pbs	Final interface evaluation, artifact hashes, PBS job execution.	Assignment-specific orchestration; course interfaces / IITD HPC [3].
tests/test_*.py scripts/smoke_test.sh .github/workflows/	Hand examples, regression/export/staff-loading checks and CI.	Course tests/scaffolding retained or extended; added checks.
configs/*.json results/*.json and evidence subfolders	Frozen search spaces, selected settings, CV/fit/runtime/hash receipts.	Own experiment configurations and machine-generated evidence; grader feedback labeled separately.
README.md docs/EXPERIMENTS*.md docs/SUBMISSION_NOTES.md	Usage, method, split declarations, frozen protocols and provenance.	Course README/docs extended; assignment-specific documentation.
requirements.txt / .gitignore reports/submission_597/	Dependencies/exclusions; retrospective report sweep, evidence and PDF generation.	Course manifests adapted; report-only code generated with Codex.

References and sources. [1] Course Assignment 3 specification, RankNet primer and starter repository; Christopher J. C. Burges, *From RankNet to LambdaRank to LambdaMART: An Overview*, MSR-TR-2010-82, 2010 (Microsoft Research). [2] LightGBM 4.6.0 parameter documentation. [3] IIT Delhi HPC guide. Recorded metrics/configurations: archived results/dev_metrics_v12.json, lambdamart_cv_v3.json, lambdamart_cv_v12.json, neural_selection_v12.json and tree_fit_v10.json; user-supplied #597 grader snapshot.

Artifact provenance. Competition archive: rank-fusion-submission-v12.zip; commit f28b61880a56cac51566890671405db013bb3c94.
SHA-256:
e42c56344bcce528e48b803ba2fb265aa3adb93cddc53791876dbd9f373b6e28
The report adds a post-submission diagnostic; the evaluated archive and artifacts remain unchanged.